



香港科技大學
THE HONG KONG
UNIVERSITY OF SCIENCE
AND TECHNOLOGY

ELEC 3200: System Modeling, Analysis and Control

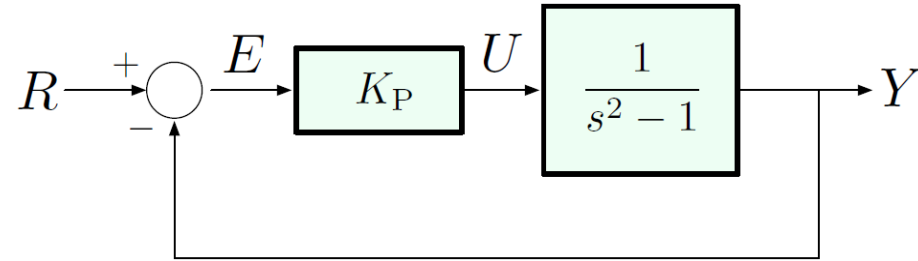
Lecture 17: PID Control and System Type

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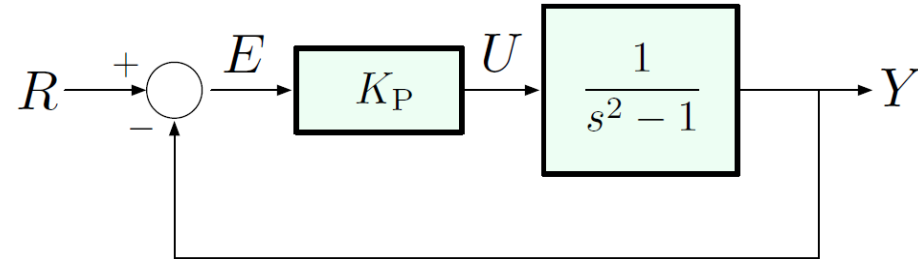
Department of Electronic and Computer Engineering

Proportional Feedback



K_P – “proportional gain” (P-gain) $U = K_P E$

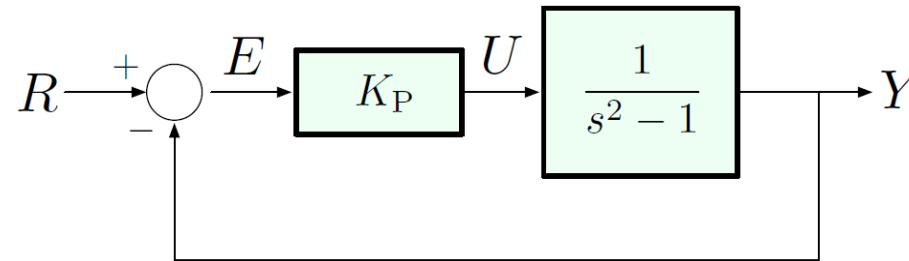
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Let's try to find a value of K_P that would stabilize the system:

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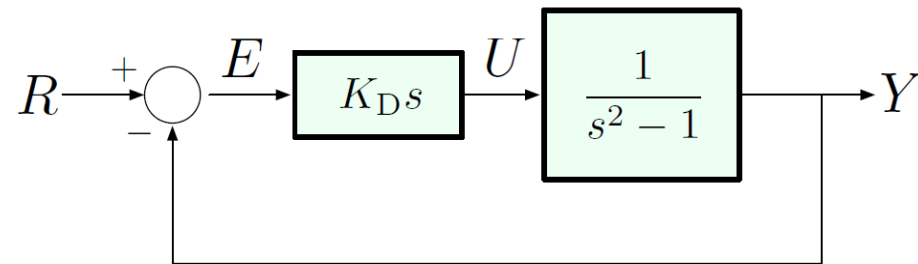
$$\frac{Y}{R} = \frac{\frac{K_P}{s^2 - 1}}{1 + \frac{K_P}{s^2 - 1}} = \frac{K_P}{s^2 - 1 + K_P}$$

— the polynomial in the denominator has zero coefficient of s
 $\implies$ necessary condition for stability is not satisfied.

The feedback system is *not stable for any value of K_P !!*

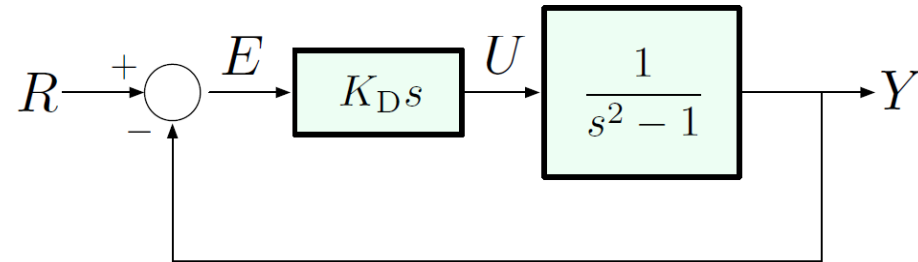
Derivative Feedback

Let's feed the *derivative of the error*, multiplied by some gain, back into the plant:



Motivation: derivative = rate of change; faster change $\implies$ more control needed.

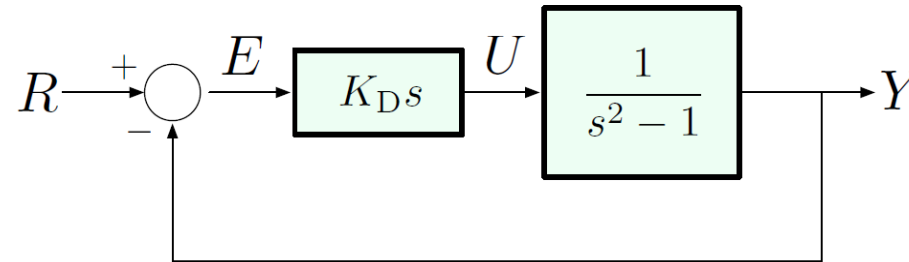
Back to Analysis: Derivative Feedback



$$\frac{Y}{R} = \frac{\frac{K_D s}{s^2 - 1}}{1 + \frac{K_D s}{s^2 - 1}} = \frac{K_D s}{s^2 + K_D s - 1}$$

— still not good: the denominator has a negative coefficient
 $\implies$ not stable; also we have picked up a zero at the origin.

Back to Analysis: Derivative Feedback



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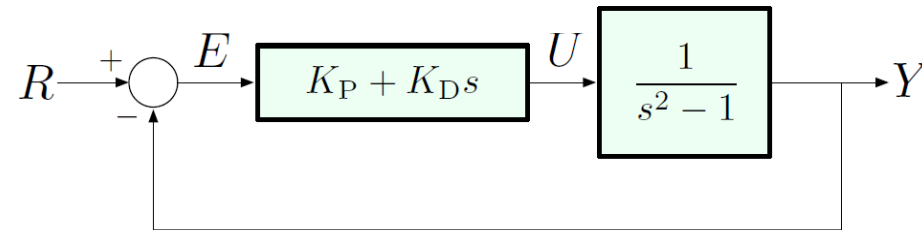
— still not good: the denominator has a negative coefficient
 $\implies$ not stable; also we have picked up a zero at the origin.

But:

- ▶ P-control affected the coefficient of s^0 (constant term)
- ▶ D-control affected the coefficient of s

— let's combine them!!

Proportional-Derivative (PD) Control



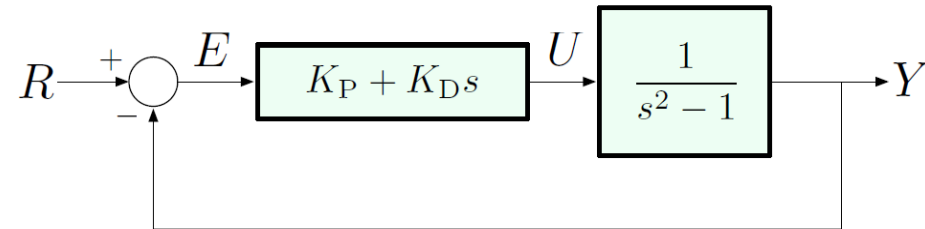
$$\frac{Y}{R} = \frac{\frac{K_P + K_D s}{s^2 - 1}}{1 + \frac{K_P + K_D s}{s^2 - 1}} = \frac{K_P + K_D s}{s^2 + K_D s + K_P - 1}$$

— now, if we set $K_D > 0$ and $K_P > 1$, then the transfer function will be stable.

Even more: by choosing K_P and K_D , we can *arbitrarily* assign coefficients of the denominator, and therefore the poles of the transfer function:

PD control gives us *arbitrary pole placement!!*

Proportional-Derivative (PD) Control



$$\frac{Y}{R} = \frac{K_P + K_D s}{s^2 + K_D s + K_P - 1}$$

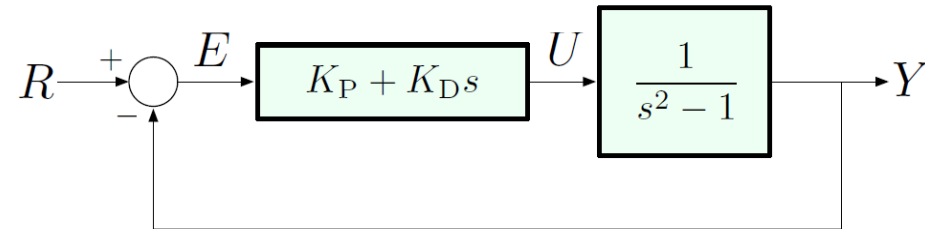
By choosing K_P, K_D , we can achieve **arbitrary pole placement!!**

Also note that the addition of P-gain moves the zero:

$$K_D s + K_P = 0 \quad \text{LHP zero at } -\frac{K_P}{K_D}$$

But what's missing?

Proportional-Derivative (PD) Control



$$\frac{Y}{R} = \frac{K_P + K_D s}{s^2 + K_D s + K_P - 1}$$

By choosing K_P, K_D , we can achieve **arbitrary pole placement!!**

Also note that the addition of P-gain moves the zero:

$$K_D s + K_P = 0 \quad \text{LHP zero at } -\frac{K_P}{K_D}$$

But what's missing? $\text{DC gain} = \left. \frac{Y}{R} \right|_{s=0} = \frac{K_P}{K_P - 1} \neq 1$

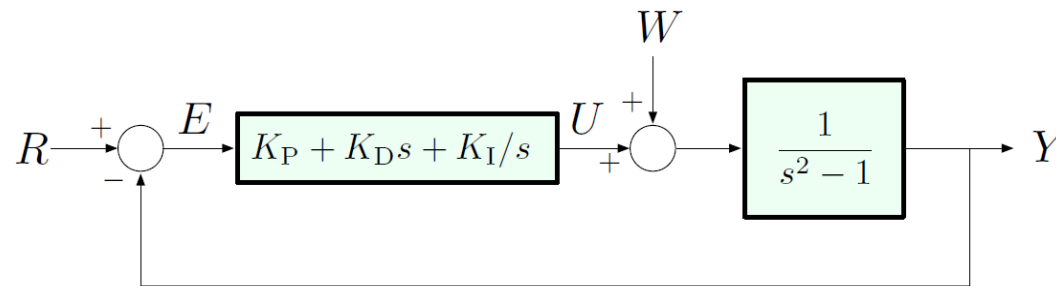
— can't have perfect tracking of constant reference.

Proportional-Integral-Derivative (PID) Control

Let us try

$$U = \left(K_P + K_D s + \frac{K_I}{s} \right) E \quad - \text{the classic three-term controller}$$

In fact, let's also throw in a constant disturbance:

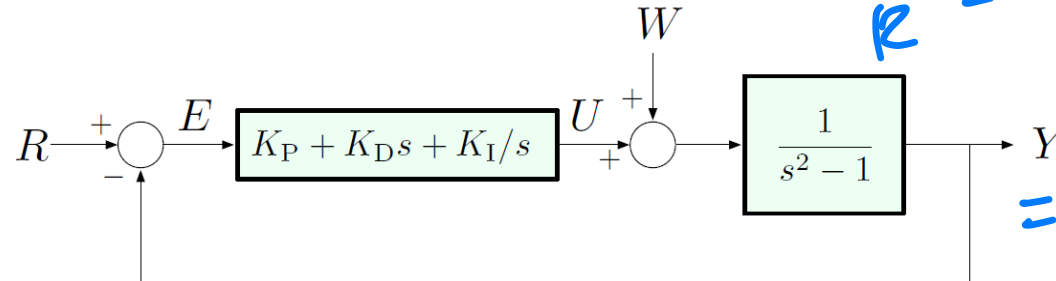


We will see that, with PID control, the goals of

- ▶ tracking a constant reference r
- ▶ rejecting a constant disturbance w

can be accomplished in one shot.

Proportional-Integral-Derivative (PID) Control



$$\frac{1}{K} = \frac{1}{1 + \frac{1}{s^2 - 1}}$$

$$= \frac{1}{s^2 - 1 + 1}$$

$$(s^3 - s + K_P s + K_D s^2 + K_I)Y = (K_P s + K_D s^2 + K_I)R + W s$$

Therefore,

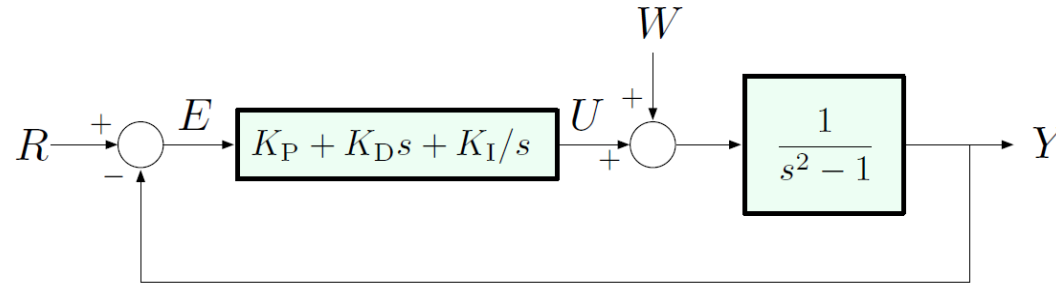
$$= \frac{K_P s + K_D s^2 + K_I}{s^3 - 1 + K_P s + K_D s^2 + \frac{K_I}{s}}$$

$$Y = \frac{K_D s^2 + K_P s + K_I}{s^3 + K_D s^2 + (K_P - 1)s + K_I} R + \frac{K_D s^2 + K_P s + K_I}{s^3 + K_D s^2 + (K_P - 1)s + K_I} \frac{W s}{s}$$

$$\begin{matrix} 1 & K_P - 1 \\ K_D & K_I \\ \frac{K_D(K_P - 1) + K_I}{K_D} & 0 \\ K_I & 0 \end{matrix}$$

$$\begin{matrix} K_D > 0 \\ K_D(K_P - 1) - K_I > 0 \\ K_D(K_P - 1) > K_I \\ K_I > 0 \end{matrix}$$

Proportional-Integral-Derivative (PID) Control

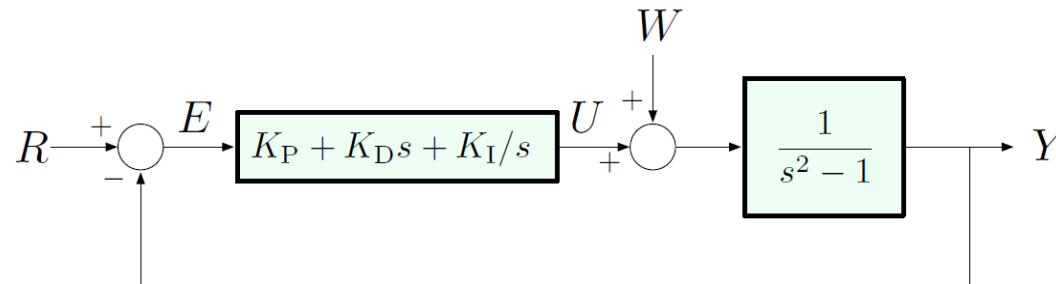


$$Y = \frac{K_D s^2 + K_P s + K_I}{s^3 + K_D s^2 + (K_P - 1)s + K_I} R + \frac{s}{s^3 + K_D s^2 + (K_P - 1)s + K_I} W$$

Stability:

- ▶ need $K_D > 0$, $K_P > 1$, $K_I > 0$ (necessary condition) and $K_D(K_P - 1) > K_I$ (Routh–Hurwitz for 3rd-order)
- ▶ can still assign coefficients arbitrarily by choosing K_P, K_I, K_D

Proportional-Integral-Derivative (PID) Control



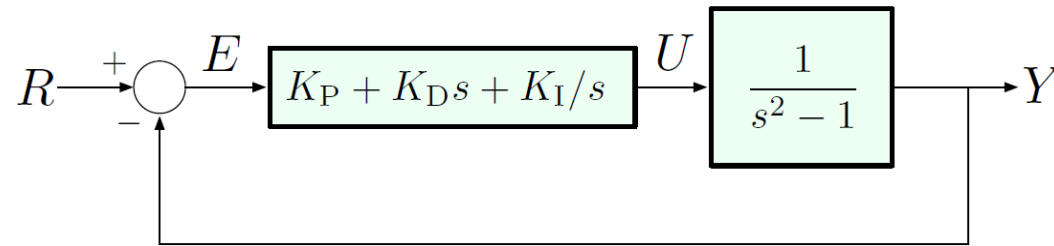
$$Y = \frac{K_D s^2 + K_P s + K_I}{s^3 + K_D s^2 + (K_P - 1)s + K_I} R + \frac{s}{s^3 + K_D s^2 + (K_P - 1)s + K_I} W$$

Reference tracking:

$$\text{DC gain}(R \rightarrow Y) = \left. \frac{K_D s^2 + K_P s + K_I}{s^3 + (K_P - 1)s + K_D s^2 + K_I} \right|_{s=0} = 1$$

— so, with the addition of I-feedback, we remove earlier limitation and achieve *perfect tracking*!

Proportional-Integral-Derivative (PID) Control



$$Y = \frac{K_D s^2 + K_P s + K_I}{s^3 + K_D s^2 + (K_P - 1)s + K_I} R + \frac{s}{s^3 + K_D s^2 + (K_P - 1)s + K_I} W$$

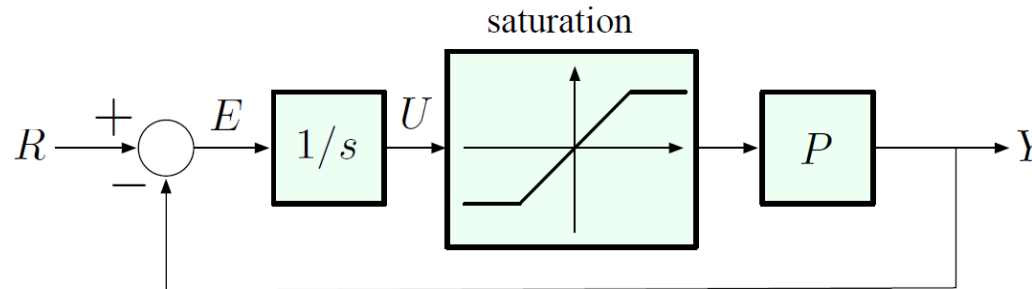
Disturbance rejection:

$$\text{DC gain}(W \rightarrow Y) = \left. \frac{s}{s^3 + (K_P - 1)s + K_D s^2 + K_I} \right|_{s=0} = 0$$

— so, integral gain also gives *complete attenuation* of *constant* disturbances!!

Wind-Up Phenomenon

For I controller



When the actuator saturates, the error continues to be integrated, resulting in large overshoot.

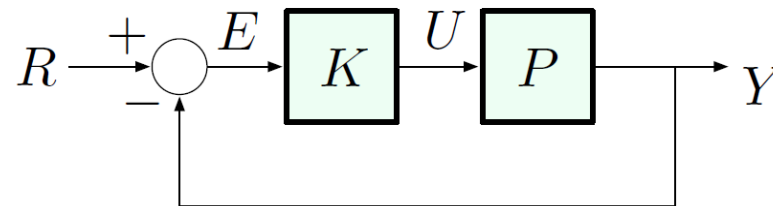
We say that the integrator “winds up:” the error may be small, but its integrated past history builds up.

There are various *anti-windup* schemes to deal with this practically important issue. (Essentially, we attempt to detect the onset of saturation and turn the integrator off.)

2nd Oscillating, need
beware of k_2 , need to $\downarrow k_1$

System Type

The fact that $1/s$ leads to perfect tracking of constant references and perfect rejection of constant disturbances is a special case of a more general analysis.

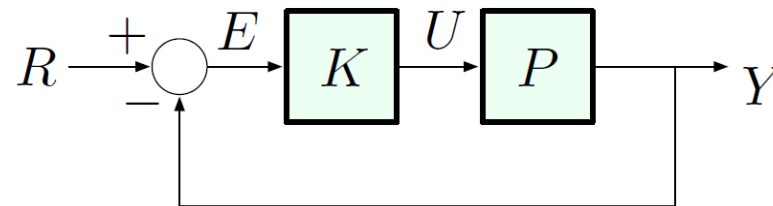


Consider the reference $r(t) = \frac{t^k}{k!}1(t) \longleftrightarrow R(s) = \frac{1}{s^{k+1}}$

Test Inputs:	$r(t) = \frac{t^k}{k!}1(t)$	$k=0$: step (position)
	$R(s) = \frac{1}{s^{k+1}}$	$k=1$: ramp (velocity)
		$k=2$: parabola (acceleration)

System Type

The fact that $1/s$ leads to perfect tracking of constant references and perfect rejection of constant disturbances is a special case of a more general analysis.



Consider the reference $r(t) = \frac{t^k}{k!}1(t) \longleftrightarrow R(s) = \frac{1}{s^{k+1}}$

Error signal: $E = \frac{1}{1 + KP} R = \frac{1}{1 + KP} \frac{1}{s^{k+1}}$

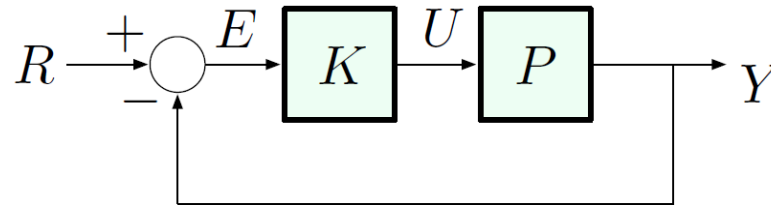
FVT gives (assuming stability):

$$e(\infty) = sE(s) \Big|_{s=0} = \frac{1}{1 + KP} \frac{1}{s^k} \Big|_{s=0}$$

— let's see how the forward gain affects tracking performance.

☆☆? System Type

how many poles $s=0$?

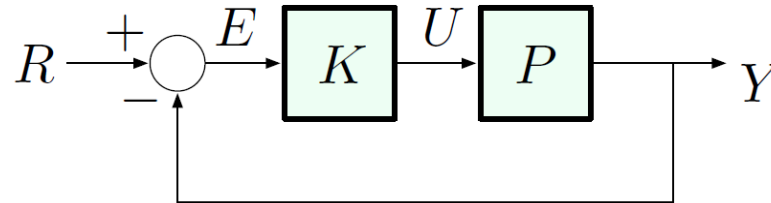


System type: the number n of poles the forward-loop transfer function KP has at the origin. It is the degree of the lowest-degree polynomial that cannot be tracked *in feedback* with zero steady-state error.

↪ proof?

Note: the system type is calculated from the *forward-loop* transfer function, although the conclusions we will draw are about the *closed-loop system*.

System Type



$$R(s) = \frac{1}{s^{k+1}} \implies E = \frac{1}{1 + KP} R = \frac{1}{1 + KP} \frac{1}{s^{k+1}}$$

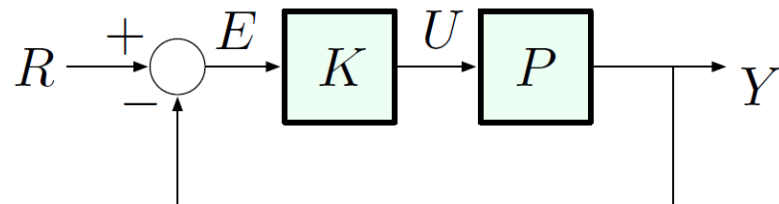
$$e(\infty) = sE(s) \Big|_{s=0} = \frac{1}{1 + KP} \frac{1}{s^k} \Big|_{s=0} \quad \text{want } e(\infty) = 0$$

— let's see how forward gain KP affects tracking performance.

Let's suppose that KP has n th-order pole at $s = 0$: $KP = \frac{K_0}{s^n}$

$$sE(s) = \frac{1}{\left(1 + \frac{K_0}{s^n}\right) s^k} = \frac{s^{n-k}}{s^n + K_0} \quad \text{— what about } sE(s) \Big|_{s=0} ?$$

System Type



Let's suppose that KP has n th-order pole at $s = 0$: $KP = \frac{K_0}{s^n}$

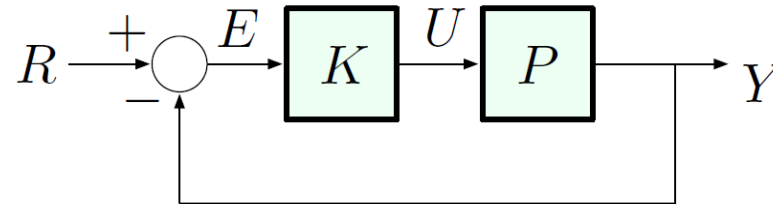
$$sE(s) = \frac{1}{\left(1 + \frac{K_0}{s^n}\right) s^k} = \frac{s^{n-k}}{s^n + K_0} \quad \text{— what about } sE(s)\Big|_{s=0}?$$

Recall: reference $r(t)$ is a polynomial of degree k

Three cases to consider —

- ▶ $n > k$: $e(\infty) = 0$ perfect tracking
- ▶ $n = k$: $e(\infty) = \text{const} \neq 0$ imperfect tracking
- ▶ $n < k$: $e(\infty) = \infty$ no tracking

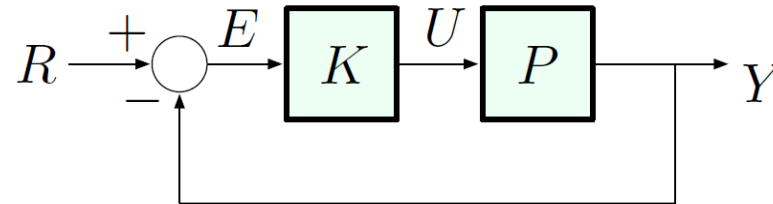
System Type: Examples



System type is the degree of the lowest-degree polynomial that cannot be tracked *in feedback* with zero steady-state error.

- ▶ **Type 0:** no pole at the origin. This is what we had without the I-gain: nonzero SS error to constant references.
- ▶ **Type 1:** a single pole at the origin. This is what we get with I-gain: can track (respectively, reject) constant references (respectively, disturbances) with zero error.
 - ▶ can check that we have a nonzero (but finite) error when tracking ramp references
- ▶ **Type 2:** a double pole at the origin. Can track ramp references without error, but not $t^2, t^3, \dots$

System Type: Examples



System type is the degree of the lowest-degree polynomial that cannot be tracked *in feedback* with zero steady-state error.

Type (n)	Disturbance (k)		
	Step ($k=0$)	Ramp ($k=1$)	Parabola ($k=2$)
Type 0	*	∞	∞
Type 1	0	*	∞
Type 2	0	0	*

$$0 < * < \infty$$

PID Control: Summary & Further Comments

P-gain simplest to implement, but not always sufficient for stabilization

K_p works very well in sensitivity!

D-gain helps achieve stability, improves time response (more control over pole locations)

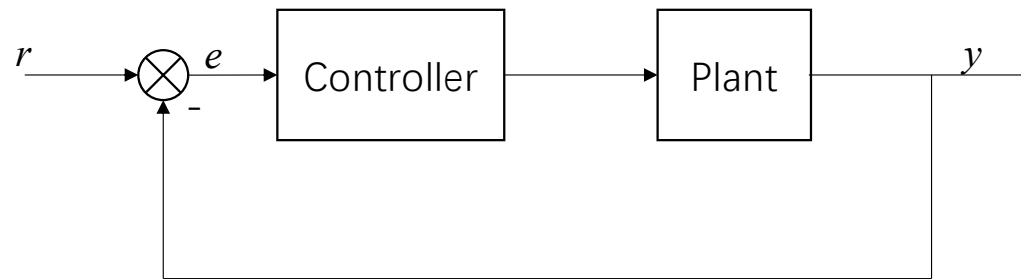
- ▶ arbitrary pole placement only valid for 2nd-order response; in general, we still have control over two *dominant poles*
- ▶ cannot be implemented directly, so need approximate implementation; D-gain also amplifies noise

I-gain essential for perfect steady-state tracking of constant reference and rejection of constant disturbance

- ▶ but $1/s$ is not a stable element by itself, so one must be careful: it can destabilize the system if the feedback loop is broken (*integrator wind-up*)

More words about the system type and PID controller

- When we design a controller, the first thing we want is to make the system stable.
- However, “stable” is defined as bounded output, so it doesn’t mean zero steady-state error.

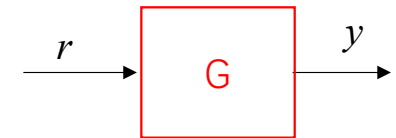
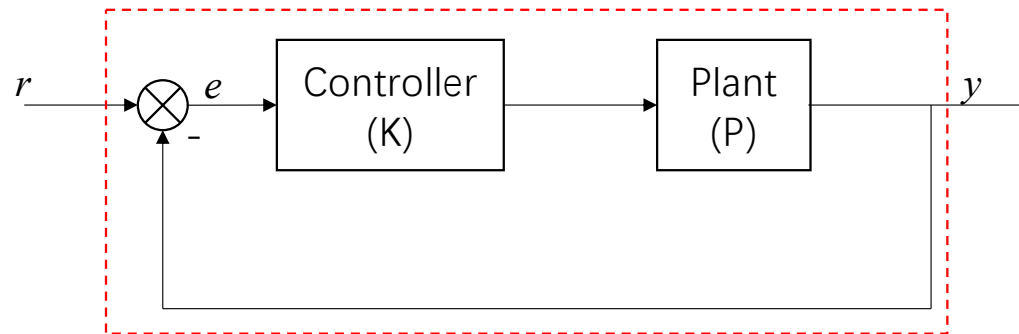


Error (e) between the output (y) and reference value (r):

$$\left\{ \begin{array}{l} e = 0 \\ e \leq |M| \end{array} \right.$$

So, the question is: how to design the controller to make $e=0$?

More words about the system type and PID controller



Input signal (r):

order (k) of
the input

$k=0$

$k=1$

$k=2$

$\vdots$

Function name	$x(t)$	$X(s)$
Unit impulse	$\delta(t)$	1
Unit step	$\sigma(t)$	$\frac{1}{s}$
Unit ramp	$t\sigma(t)$	$\frac{1}{s^2}$
Unit parabola	$\frac{t^2}{2}\sigma(t)$	$\frac{1}{s^3}$
n th-order ramp	$\frac{t^n}{n!}\sigma(t)$	$\frac{1}{s^{n+1}}$

Let's consider a special case, i.e., the input is unit step. What's the output?

$$Y(s) = R(s) * G(s) = 1/s * G(s)$$



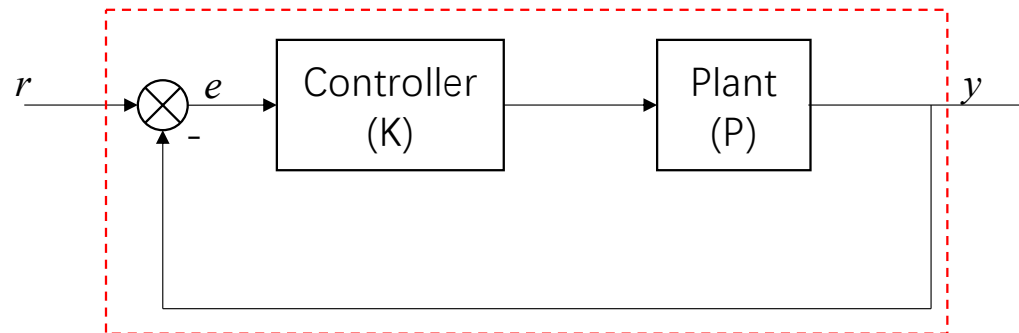
$$y(\infty) = s * Y(s)|_{s=0} = s * 1/s * G(s)|_{s=0} = G(0)$$

----- FVT Theorem

(This is the definition of DC gain!)

Hint: you can see, the order of s is $k+1$

More words about the system type and PID controller



Input signal (r):

order (k) of
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$k=0$

$k=1$

$k=2$

$\vdots$

Function name	$x(t)$	$X(s)$
Unit impulse	$\delta(t)$	1
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Unit parabola	$\frac{t^2}{2}\sigma(t)$	$\frac{1}{s^3}$
n th-order ramp	$\frac{t^n}{n!}\sigma(t)$	$\frac{1}{s^{n+1}}$



Hint: you can see, the order of s is $k+1$

Let's consider a general case, what's the error e ?

$$E(s) = \frac{1}{1 + KP} R(s)$$

$$e(\infty) = sE(s)|_{s=0} = s \frac{1}{1 + KP} R(s)|_{s=0} \quad - \text{FVT Theorem}$$

$$= s \frac{1}{1 + KP} \frac{1}{s^{k+1}} |_{s=0}$$

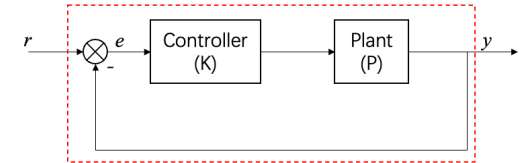
$$= \frac{1}{1 + KP} \frac{1}{s^k} |_{s=0}$$

More words about the system type and PID controller

Our purpose is to make $e(\infty) = 0$, therefore, the follow equation should hold:

$$e(\infty) = \frac{1}{1 + KP} \frac{1}{s^k} \Big|_{s=0} = 0$$

input, k-th order



Note: KP is the open-loop transfer function.

Now, let's consider the value of KP. Suppose the number of **poles at the origin ($s = 0$) of KP** is **n**. So, KP can be represented as:

$$KP = \frac{(s + z_1)(s + z_2) \dots}{s^n (s + p_1)(s + p_2) \dots}$$

So, we have:

$$\begin{aligned} \lim_{s \rightarrow 0} \frac{1}{1 + KP} \frac{1}{s^k} &= \lim_{s \rightarrow 0} \frac{1}{\left(1 + \frac{(s + z_1)(s + z_2) \dots}{s^n (s + p_1)(s + p_2) \dots}\right) s^k} \\ &= \frac{s^{n-k}}{s^n + \frac{(0 + z_1)(0 + z_2) \dots}{(0 + p_1)(0 + p_2) \dots}} \Big|_{s=0} \end{aligned}$$

Now you can see that we only need to compare **n** and **k**:

n > k	$e(\infty) = 0$
n = k	$e(\infty) = \text{Constant value}$
n < k	$e(\infty) = \infty$

More words about the system type and PID controller

If we represent it in a table:

Type (n)	Input Disturbance (k)			
	Step ($k=0$)	Ramp ($k=1$)	Parabola ($k=2$)	
Type 0	*	∞	∞	$\left\{ \begin{array}{ll} n > k & e(\infty) = 0 \\ n = k & e(\infty) = * \\ n < k & e(\infty) = \infty \end{array} \right.$
Type 1	0	*	∞	
Type 2	0	0	*	

$0 < * < \infty$

The definition of System Type:

- The number n of poles the forward-loop transfer function K_P has at the origin.
- It is the degree of the lowest-degree polynomial that cannot be tracked in feedback with zero steady-state error.

$$Y(s) = \frac{1}{1 + K_P} R(s)$$

More words about the system type and PID controller

Example & Exercise:

$$E(s) = \frac{1}{1 + K_P} R(s)$$

$$e(\infty) = \lim_{s \rightarrow 0} \frac{s}{1 + K_P} R(s)$$

A unity negative feedback control system has the following open-loop transfer function:

$$G(s) = \frac{10}{s(s+4)}$$

$$\frac{s}{1 + \frac{10}{s(s+4)}} = \frac{s - s(1+4)}{s(s+4) + 10} \cdot \frac{1}{s} = 0$$

Tasks:

1. Determine the **system type**.

second order system, type 1

2. Find the **steady-state error** for the following standard inputs:

• (a) Unit step input: $r(t) = 1$

$$\frac{1}{s} \Rightarrow \lim_{s \rightarrow 0} s \frac{1}{1 + K_P} = \frac{10}{s(s+4)} \Big|_{s=0} = 0$$

• (b) Unit ramp input: $r(t) = t$

$$\frac{1}{s^2} \Rightarrow \lim_{s \rightarrow 0} s \frac{1}{1 + K_P} = \frac{10}{s(s+4)} \Big|_{s=0} = 0.4$$

• (c) Unit parabolic input: $r(t) = \frac{t^2}{2}$

$$\frac{1}{s^3} \Rightarrow \lim_{s \rightarrow 0} s \frac{1}{1 + K_P} = \frac{10}{s(s+4)} \Big|_{s=0} = \infty$$

degree = 1

More words about the system type and PID controller

Example & Exercise:

A unity negative feedback control system has the following open-loop transfer function:

$$G(s) = \frac{10}{s(s + 4)}$$

1. Determine the System Type

- System Type is determined by the number of **poles at the origin ($s = 0$)** in the open-loop transfer function.
- Here, $G(s)$ has one pole at $s = 0$.

✅ Answer: Type 1 system

More words about the system type and PID controller

Example & Exercise:

A unity negative feedback control system has the following open-loop transfer function:

$$G(s) = \frac{10}{s(s+4)}$$

2. Find the **steady-state error** for the following standard inputs:

- (a) Unit step input: $r(t) = 1$ $\longrightarrow$ $\frac{1}{s}$
 - (b) Unit ramp input: $r(t) = t$ $\longrightarrow$ $\frac{1}{s^2}$
 - (c) Unit parabolic input: $r(t) = \frac{t^2}{2}$ $\longrightarrow$ $\frac{1}{s^3}$
- $$e(\infty) = sE(s)|_{s=0} = s \frac{1}{1+KP} R(s)|_{s=0}$$

Input Type	Steady-State Error e_{ss}
Step $r(t) = 1$	0
Ramp $r(t) = t$	0.4
Parabolic $r(t) = \frac{t^2}{2}$	∞

More words about the system type and PID controller

Think about this question: Why PID controller can help?

Answer: The PID controller can effectively increase the system type!

A **PID controller** can turn a **Type 0 system** (i.e., one with **no integrator**) into an **effective Type 1 or higher system** by introducing an **integrator (the “I” term)** in the control loop. This allows the system to **achieve perfect steady-state tracking** of step inputs—or even higher-order inputs depending on the system design.

Example:

Suppose we have a system: $G(s) = \frac{1}{s+1}$ - Type 0 system

Let's design a simple PI controller: $C(s) = 1 + \frac{1}{s}$

The new system will be: $G_{\text{open}}(s) = \left(1 + \frac{1}{s}\right) \cdot \frac{1}{s+1} = \frac{s+1}{s(s+1)} = \frac{1}{s}$ - Type 1 system !

More words about the system type and PID controller

Discussion: Why do we only consider the poles at the origin ($s=0$)? How about other poles?

$$e(\infty) = sE(s)|_{s=0} = s \frac{1}{1 + KP} R(s)|_{s=0} \quad KP = \frac{(s + z_1)(s + z_2) \dots}{s^n (s + p_1)(s + p_2) \dots}$$

Answer:

- **System type** is determined solely by the number of **poles at the origin ($s = 0$)** in the open-loop transfer function $G(s)$ —that is, the number of **integrators** in the system.
- Other poles (those not at the origin) affect the system's **transient response** (like speed, overshoot, etc.), but they do **not influence the system type or steady-state error behavior** for standard inputs such as step, ramp, or parabolic.

We will discuss the effect of other poles and zeros to the system in the coming lectures!

Review

- What's the meaning of P, I and D?
- Can PD control give arbitrary pole placement? Why?
- What's the limitation of PD control? How to address it?
- What's wind-up phenomenon?
- What's the meaning of system type?